Agent1: " To simplify the expression \\((-k + 4) + (-2 + 3k)\\), we can follow these steps:\n\n### Step 1: Remove the Parentheses\nFirst, remove the parentheses by distributing the addition sign. This gives us:\n\\[\n(-k + 4) + (-2 + 3k) = -k + 4 - 2 + 3k\n\\]\n\n### Step 2: Combine Like Terms\nNext, combine the like terms. The like terms in this expression are:\n- The terms involving \\(k\\): \\(-k\\) and \\(3k\\)\n- The constant terms: \\(4\\) and \\(-2\\)\n\nCombine the \\(k\\) terms:\n\\[\n-k + 3k = 2k\n\\]\n\nCombine the constant terms:\n\\[\n4 - 2 = 2\n\\]\n\n### Step 3: Write the Simplified Expression\nNow, put the combined terms together:\n\\[\n2k + 2\n\\]\n\nSo, the simplified form of the expression \\((-k + 4) + (-2 + 3k)\\) is:\n\\[\n\\boxed{2k + 2}\n\\]",

Agent2: " To simplify the expression \\((-k + 4) + (-2 + 3k)\\), we will follow these steps:\n\n### Step 1: Remove the Parentheses\nFirst, remove the parentheses by distributing the addition sign:\n\n\\[\n(-k + 4) + (-2 + 3k) = -k + 4 - 2 + 3k\n\\]\n\n### Step 2: Combine Like Terms\nNext, combine the like terms. Like terms are those that contain the same variable or are constants.\n\n- **Combine the \\(k \\)-terms:**\n \n \\[\n -k + 3k\n \\]\n\n- **Combine the constant terms:**\n \n \\[\n 4 - 2\n \\]\n\n### Step 3: Perform the Arithmetic\nCalculate each part separately.\n\n1. For the \\(k \\)-terms:\n \n \\[\n -k + 3k = 2k\n \\]\n\n2. For the constant terms:\n \n \\[\n 4 - 2 = 2\n \\]\n\n### Step 4: Write the Simplified Expression\nNow, put the combined terms together:\n\n\\[\n2k + 2\n\\]\n\n### Final Answer\nThus, the simplified form of the expression is:\n\n\\[\n\\boxed{2k + 2}\n\\]",

Agent3: " What is the coefficient of k?\n\nAlright, I have this math problem to solve:\n\n**Simplify \\((-k + 4) + (-2 + 3k)\\). What is the coefficient of \\(k\\)?**\n\nAt first glance, it seems like I need to combine like terms to simplify the expression. Let's break it down step by step.\n\n### Step 1: Understand the Expression\n\nThe expression is \\((-k + 4) + (-2 + 3k)\\). I need to simplify this by combining like terms. \n\n**What are like terms?**\n\nLike terms are terms that have the same variable raised to the same power. In this case, \\(-k\\) and \\(3k\\) are like terms because they both have the variable \\(k\\) raised to the first power. Similarly, \\(4\\) and \\(-2\\) are like terms because they are both constants (they don't have any \\(k\\) in them).\n\n### Step 2: Remove the Parentheses\n\nFirst, I'll remove the parentheses to make it easier to combine the terms.\n\n\\[\n(-k + 4) + (-2 + 3k)\n\\]\n\nRemoving the parentheses (and noting that adding \\(-2\\) is the same as subtracting \\(2\\)):\n\n\\[\n-k + 4 - 2 + 3k\n\\]\n\nWait, is that correct? Let's double-check.\n\nActually, the original expression is:\n\n\\[\n(-k + 4) + (-2 + 3k)\n\\]\n\nSo, I'm adding two groups:\n\n1. \\(-k + 4\\)\n2. \\(-2 + 3k\\)\n\nWhen I add them together, it's:\n\n\\[\n(-k + 4) + (-2 + 3k) = -k + 4 - 2 + 3k\n\\]\n\nYes, that makes sense. I'm just adding the two parts together.\n\n### Step 3: Combine Like Terms\n\nNow, I'll combine the \\(k\\) terms and the constant terms.\n\n**Combining the \\(k\\) terms:**\n\n\\[\n-k + 3k\n\\]\n\n\\[\n(-1 + 3)k = 2k\n\\]\n\n**Combining the constant terms:**\n\n\\[\n4 - 2\n\\]\n\n\\[\n2\n\\]\n\nSo, putting it all together:\n\n\\[\n2k + 2\n\\]\n\n### Step 4: Identify the Coefficient of \\(k\\)\n\nThe simplified expression is \\(2k + 2\\). The coefficient of \\(k\\) is the number that \\(k\\) is multiplied by. In this case, that's \\(2\\).\n\n### Verification\n\nLet's verify each step to make sure I didn't make any mistakes.\n\n1. **Original Expression:**\n\n \\[\n (-k + 4) + (-2 + 3k)\n \\]\n\n2. **Removing Parentheses:**\n\n \\[\n -k + 4 - 2 + 3k\n \\]\n\n - \\(-k\\) from the first parentheses.\n - \\(+4\\) from the first parentheses.\n - \\(+(-2)\\) from the second parentheses (which is \\(-2\\)).\n - \\(+3k\\) from the second parentheses.\n\n So, combining them:\n\n \\[\n -k + 4 - 2 + 3k\n \\]\n\n3. **Combining Like Terms:**\n\n - \\(-k + 3k = 2k \\)\n - \\(4 - 2 = 2 \\)\n\n Resulting in:\n\n \\[\n 2k + 2\n \\]\n\nEverything checks out. The coefficient of \\(k\\) is indeed \\(2\\).\n\n### Final Answer\n\nThe coefficient of \\(k\\) in the simplified expression is:\n\n\\[\n\\boxed{2}\n\\]",

Agent4: " To simplify the expression \\((-k + 4) + (-2 + 3k)\\), let's follow the steps carefully.\n\n### Step-by-Step Simplification\n\n1. **Remove the Parentheses:**\n \\[\n (-k + 4) + (-2 + 3k) = -k + 4 - 2 + 3k\n \\]\n\n2. **Combine Like Terms:**\n - **Combine the \\(k \\)-terms:**\n \\[\n -k + 3k = 2k\n \\]\n - **Combine the constant terms:**\n \\[\n 4 - 2 = 2\n \\]\n\n3. **Write the Simplified Expression:**\n \\[\n 2k + 2\n \\]\n\nTherefore, the simplified form of the expression \\((-k + 4) + (-2 + 3k)\\) is:\n\n\\[\n\\boxed{2k + 2}\n\\]",

Agent5: " The coefficient of k is\n\nAlright, I have this math problem to solve:\n\n**Simplify \\((-k + 4) + (-2 + 3k)\\). The coefficient of \\(k\\) is?**\n\nFirst, I need to simplify the expression \\((-k + 4) + (-2 + 3k)\\). After simplifying, I should find the coefficient of \\(k\\), which is the number that \\(k\\) is multiplied by.\n\n### Step 1: Remove the Parentheses\n\nI see that the expression has parentheses. To simplify, I should first remove the parentheses by distributing the addition operation. However, since it's just a simple addition of two expressions inside parentheses, I can directly combine them.\n\nSo, \\((-k + 4) + (-2 + 3k)\\) becomes:\n\n\\[\n-k + 4 - 2 + 3k\n\\]\n\nWait, is that correct? Let's think about it.\n\n### Understanding the Addition\n\nThe original expression is:\n\n\\[\n(-k + 4) + (-2 + 3k)\n\\]\n\nThis means I'm adding two parts together: \\((-k + 4)\\) and \\((-2 + 3k)\\).\n\nTo combine them, I can write:\n\n\\[\n(-k) + 4 + (-2) + 3k\n\\]\n\nWhich simplifies to:\n\n\\[\n-k + 4 - 2 + 3k\n\\]\n\nYes, that makes sense. I'm adding all the terms together.\n\n### Step 2: Combine Like Terms\n\nNext, I need to combine the like terms. Like terms are terms that have the same variable part.\n\nIn this expression, I have:\n\n- \\(-k\\) and \\(3k\\) (both are terms with \\(k\\))\n- \\(4\\) and \\(-2\\) (both are constant terms)\n\nSo, I'll combine the \\(k\\) terms and the constant terms separately.\n\n**Combining the \\(k\\) terms:**\n\n\\[\n-k + 3k\n\\]\n\n\\[\n= (-1 + 3)k\n\\]\n\n\\[\n= 2k\n\\]\n\n**Combining the constant terms:**\n\n\\[\n4 - 2\n\\]\n\n\\[\n= 2\n\\]\n\n### Step 3: Write the Simplified Expression\n\nNow, I'll put the combined terms together:\n\n\\[\n2k + 2\n\\]\n\nSo, the simplified form of \\((-k + 4) + (-2 + 3k)\\) is \\(2k + 2\\).\n\n### Step 4: Identify the Coefficient of \\(k\\)\n\nThe problem asks for the coefficient of \\(k\\). In the expression \\(2k + 2\\), the term with \\(k\\) is \\(2k\\), and the coefficient is the number multiplied by \\(k\\), which is **2**.\n\n### Verification\n\nTo make sure I did this correctly, I'll double-check each step.\n\n1. **Removing the Parentheses:**\n\n \\[\n (-k + 4) + (-2 + 3k) = -k + 4 - 2 + 3k\n \\]\n\n Yes, that's correct.\n\n2. **Combining Like Terms:**\n\n - For \\(k\\) terms:\n \\[\n -k + 3k = 2k\n \\]\n - For constant terms:\n \\[\n 4 - 2 = 2\n \\]\n\n Both are correct.\n\n3. **Final Expression:**\n\n \\[\n 2k + 2\n \\]\n\n This matches my earlier result.\n\n4. **Coefficient of \\(k\\):**\n\n In \\(2k\\), the coefficient is 2.\n\nEverything checks out.\n\n### Final Answer\n\nThe coefficient of \\(k\\) is \\(\\boxed{2}\\)."

Figure 6. Responses after one round debate on MATH500 with 5 agents.

Critique-augmented update and constant mass. The belief update is (Definition 4.1)

$$\alpha_{i,t} = \alpha_{i,t-1} + \beta_{i,t-1} + w_i c_{i,t}, \qquad \beta_{i,t-1} \in \mathbb{R}_+^K, \quad w_i \geq 0. \tag{16}$$

Throughout the proofs we use the constant-mass assumption

$$\|\beta_{i,t}\|_1 = m_\beta, \qquad \text{for all } i, t, \tag{17}$$

which corresponds to adding a fixed "budget" of private pseudo-observations per round.

Why assume constant $\|\beta\|_1$. We interpret $\beta_{i,t}$ as a *fixed-budget* private signal extracted from the debate context (*e.g.,* a bounded scoring head or a fixed number of self-critique samples), hence its total pseudo-count mass is controlled. Assuming $\|\beta_{i,t}\|_1 = m_\beta$ keeps the per-round update magnitude comparable and yields closed-form drift expressions. The analysis extends to time-varying masses by replacing m_β with $\|\beta_{i,t}\|_1$, at the cost of heavier notation.

Filtration. Let $\mathcal{F}_t$ be the σ-algebra of information *available at the start of round* t (before sampling the round-t responses $y_{\cdot,t}$). Since $\beta_{\cdot,t-1}$ is computed from the round-$t-1$ debate context (which includes $y_{\cdot,t-1}$), we include β only up to time $t-1$:

$$\mathcal{F}_t := \sigma\Big(\{\alpha_{\cdot,s}\}_{s\leq t}, \{y_{\cdot,s}\}_{s\leq t-1}, \{\beta_{\cdot,s}\}_{s\leq t-1}\Big).$$

Equivalently, $\mathcal{F}_{t-1} = \sigma(\{\alpha_{\cdot,s}\}_{s\leq t-1}, \{y_{\cdot,s}\}_{s\leq t-2}, \{\beta_{\cdot,s}\}_{s\leq t-2})$, so conditioning on $\mathcal{F}_{t-1}$ treats $\alpha_{\cdot,t-1}$ as known but keeps $y_{\cdot,t-1}$ (and hence $c_{i,t}$) random. All conditional expectations below are taken with respect to $\mathcal{F}_t$.

Agent1: " What is the coefficient of k?\nAlright, I have this math problem to solve:\n\n**Simplify \\((-k + 4) + (-2 + 3k)\\). What is the coefficient of \\(k\\)?**\n\nFirst, I need to simplify the expression \\((-k + 4) + (-2 + 3k)\\). After simplifying, I should find the coefficient of \\(k\\), which is the number that \\(k\\) is multiplied by.\n\n### Step 1: Remove the Parentheses\n\nI see that the expression has parentheses. To simplify, I should first remove the parentheses by distributing the addition operation. However, since it's just a simple addition of two expressions inside parentheses, I can directly combine them.\n\nSo, \\((-k + 4) + (-2 + 3k)\\) becomes:\n\n\\[\n-k + 4 - 2 + 3k\n\\]\n\nWait, is that correct? Let's think about it.\n\n### Understanding the Addition\n\nThe original expression is:\n\n\\[\n(-k + 4) + (-2 + 3k)\n\\]\n\nThis means I'm adding two parts together: \\((-k + 4)\\) and \\((-2 + 3k)\\).\n\nTo combine them, I can write:\n\n\\[\n(-k) + 4 + (-2) + 3k\n\\]\n\nWhich simplifies to:\n\n\\[\n-k + 4 - 2 + 3k\n\\]\n\nYes, that makes sense. I'm adding all the terms together.\n\n### Step 2: Combine Like Terms\n\nNext, I need to combine the like terms. Like terms are terms that have the same variable part.\n\nIn this expression, I have:\n\n- \\(-k\\) and \\(3k\\) (both are terms with \\(k\\))\n- \\(4\\) and \\(-2\\) (both are constant terms)\n\nSo, I'll combine the \\(k\\) terms and the constant terms separately.\n\n**Combining the \\(k\\) terms:**\n\n\\[\n-k + 3k\n\\]\n\n\\[\n= (-1 + 3)k\n\\]\n\n\\[\n= 2k\n\\]\n\n**Combining the constant terms:**\n\n\\[\n4 - 2\n\\]\n\n\\[\n= 2\n\\]\n\n### Step 3: Write the Simplified Expression\n\nNow, I'll put the combined terms together:\n\n\\[\n2k + 2\n\\]\n\nSo, the simplified form of \\((-k + 4) + (-2 + 3k)\\) is \\(2k + 2\\).\n\n### Step 4: Identify the Coefficient of \\(k\\)\n\nThe problem asks for the coefficient of \\(k\\). In the expression \\(2k + 2\\), the term with \\(k\\) is \\(2k\\), and the coefficient is the number multiplied by \\(k\\), which is **2**.\n\n### Verification\n\nTo make sure I did this correctly, I'll double-check each step.\n\n1. **Removing the Parentheses:**\n\n \\[\n (-k + 4) + (-2 + 3k) = -k + 4 - 2 + 3k\n \\]\n\n Yes, that's correct.\n\n2. **Combining Like Terms:**\n\n - For \\(k\\) terms:\n \n \\[\n -k + 3k = 2k\n \\]\n \n - For constant terms:\n \n \\[\n 4 - 2 = 2\n \\]\n \n Both are correct.\n\n3. **Final Expression:**\n\n \\[\n 2k + 2\n \\]\n\n This matches my earlier result.\n\n4. **Coefficient of \\(k\\):**\n\n In \\(2k\\), the coefficient is 2.\n\nEverything checks out.\n\n### Final Answer\n\nThe coefficient of \\(k\\) is \\(\\boxed{2}\\).",

Agent2: " What is the coefficient of k?\nAlright, I have this math problem to solve:\n\n**Simplify \\((-k + 4) + (-2 + 3k)\\). What is the coefficient of \\(k\\)?**\n\nFirst, I need to simplify the expression \\((-k + 4) + (-2 + 3k)\\) and then find the coefficient of \\(k\\) in the simplified form. Let's break it down step by step.\n\n### Step 1: Understand the Expression\n\nThe expression is \\((-k + 4) + (-2 + 3k)\\). I need to simplify this by combining like terms. Like terms are terms that have the same variable raised to the same power. In this case, \\(-k\\) and \\(3k\\) are like terms, and \\(4\\) and \\(-2\\) are like terms.\n\n### Step 2: Remove the Parentheses\n\nFirst, I'll remove the parentheses to make it easier to combine the terms. The expression is:\n\n\\[\n(-k + 4) + (-2 + 3k)\n\\]\n\nNow I remove the parentheses, I get:\n\n\\[\n-k + 4 - 2 + 3k\n\\]\n\nWait, is that correct? Let's double-check.\n\nThe original expression is the sum of two parts: \\((-k + 4)\\) and \\((-2 + 3k)\\). When I add them, it's like saying:\n\n\\[\n(-k + 4) + (-2 + 3k) = -k + 4 - 2 + 3k\n\\]\n\nYes, that makes sense. I'm just adding the two parts together.\n\n### Step 3: Combine Like Terms\n\nNow, I'll combine the like terms. The like terms here are:\n\n- The \\(k \\)-terms: \\(-k\\) and \\(3k\\)\n- The constant terms: \\(4\\) and \\(-2\\)\n\n**Combining the \\(k \\)-terms:**\n\n\\[\n-k + 3k\n\\]\n\nI can combine these by adding the coefficients:\n\n\\[\n-1 + 3 = 2\n\\]\n\nSo, \\(-k + 3k = 2k\\)\n\n**Combining the constant terms:**\n\n\\[\n4 - 2\n\\]\n\n\\[\n= 2\n\\]\n\n### Step 4: Write the Simplified Expression\n\nNow, putting it all together:\n\n\\[\n2k + 2\n\\]\n\nSo, the simplified form of the expression \\((-k + 4) + (-2 + 3k)\\) is \\(2k + 2\\).\n\n### Step 5: Identify the Coefficient of \\(k\\)\n\nThe problem asks for the coefficient of \\(k\\). In the expression \\(2k + 2\\), the term with \\(k\\) is \\(2k\\), and the coefficient is the number that \\(k\\) is multiplied by. Here, that number is \\(2\\).\n\n### Verification\n\nLet's verify each step to ensure I didn't make any mistakes.\n\n1. **Original Expression:**\n\n \\[\n (-k + 4) + (-2 + 3k)\n \\]\n\n2. **Removing Parentheses:**\n\n \\[\n -k + 4 - 2 + 3k\n \\]\n\n - From \\((-k + 4)\\), I get \\(-k + 4\\).\n - From \\((-2 + 3k)\\), I get \\(-2 + 3k\\).\n - Adding them together: \\(-k + 4 - 2 + 3k\\). Yes, that's correct.\n\n3. **Combining Like Terms:**\n\n - \\(-k + 3k = 2k \\)\n - \\(4 - 2 = 2 \\)\n\n So, \\(2k + 2 \\). This matches.\n\n4. **Coefficient of \\(k\\):**\n\n In \\(2k \\), the coefficient is indeed \\(2 \\).\n\nEverything checks out. The coefficient of \\(k\\) is \\(2\\).\n\n### Final Answer\n\nThe coefficient of \\(k\\) is \\(\\boxed{2}\\).",

Agent3: " What is the coefficient of k?\n\nAlright, I have this math problem to solve:\n\n**Simplify \\((-k + 4) + (-2 + 3k)\\). What is the coefficient of \\(k\\)?**\n\nAt first glance, it seems like I need to combine like terms to simplify the expression. Let's break it down step by step.\n\n### Step 1: Understand the Expression\n\nThe expression is \\((-k + 4) + (-2 + 3k)\\). I need to simplify this by combining like terms. \n\n**What are like terms?**\n\nLike terms are terms that have the same variable raised to the same power. In this case, \\(-k\\) and \\(3k\\) are like terms because they both have the variable \\(k\\) raised to the first power. Similarly, \\(4\\) and \\(-2\\) are like terms because they are both constants (they don't have any \\(k\\) in them).\n\n### Step 2: Remove the Parentheses\n\nFirst, I'll remove the parentheses to make it easier to combine the terms.\n\n\\[\n(-k + 4) + (-2 + 3k)\n\\]\n\nRemoving the parentheses (and noting that adding \\(-2\\) is the same as subtracting \\(2\\)):\n\n\\[\n-k + 4 - 2 + 3k\n\\]\n\nWait, is that correct? Let's double-check.\n\nActually, the original expression is:\n\n\\[\n(-k + 4) + (-2 + 3k)\n\\]\n\nSo, I'm adding two groups:\n\n1. \\(-k + 4\\)\n2. \\(-2 + 3k\\)\n\nWhen I add them together, it's:\n\n\\[\n(-k + 4) + (-2 + 3k) = -k + 4 - 2 + 3k\n\\]\n\nYes, that makes sense. I'm just adding the two parts together.\n\n### Step 3: Combine Like Terms\n\nNow, I'll combine the \\(k\\) terms and the constant terms.\n\n**Combining the \\(k\\) terms:**\n\n\\[\n-k + 3k\n\\]\n\n\\[\n(-1 + 3)k = 2k\n\\]\n\n**Combining the constant terms:**\n\n\\[\n4 - 2\n\\]\n\n\\[\n2\n\\]\n\nSo, putting it all together:\n\n\\[\n2k + 2\n\\]\n\n### Step 4: Identify the Coefficient of \\(k\\)\n\nThe simplified expression is \\(2k + 2\\). The coefficient of \\(k\\) is the number that \\(k\\) is multiplied by. In this case, that's \\(2\\).\n\n### Verification\n\nLet's verify each step to make sure I didn't make any mistakes.\n\n1. **Original Expression:**\n\n \\[\n (-k + 4) + (-2 + 3k)\n \\]\n\n2. **Removing Parentheses:**\n\n \\[\n -k + 4 - 2 + 3k\n \\]\n\n - \\(-k\\) from the first parentheses.\n - \\(+4\\) from the first parentheses.\n - \\(+(-2)\\) from the second parentheses (which is \\(-2\\)).\n - \\(+3k\\) from the second parentheses.\n\n So, combining them:\n\n \\[\n -k + 4 - 2 + 3k\n \\]\n\n3. **Combining Like Terms:**\n\n - \\(-k + 3k = 2k \\)\n - \\(4 - 2 = 2 \\)\n\n Resulting in:\n\n \\[\n 2k + 2\n \\]\n\nEverything checks out. The coefficient of \\(k\\) is indeed \\(2\\).\n\n### Final Answer\n\nThe coefficient of \\(k\\) in the simplified expression is:\n\n\\[\n\\boxed{2}\n\\]",

Agent4: " What is the coefficient of k in the simplified expression?\n\nTo solve the problem, we need to simplify the expression \\((-k + 4) + (-2 + 3k)\\) and then identify the coefficient of \\(k \\) in the simplified form.\n\n### Step 1: Remove the Parentheses\n\nFirst, I'll remove the parentheses by distributing the addition sign:\n\n\\[\n(-k + 4) + (-2 + 3k) = -k + 4 - 2 + 3k\n\\]\n\n### Step 2: Combine Like Terms\n\nNext, I'll combine the like terms. The like terms in this expression are:\n\n- The terms involving \\(k \\): \\(-k\\) and \\(3k \\)\n- The constant terms: \\(4 \\) and \\(-2 \\)\n\n**Combining the \\(k \\)-terms:**\n\n\\[\n-k + 3k\n\\]\n\n\\[\n(-1 + 3)k\n\\]\n\n\\[\n2k\n\\]\n\n**Combining the constant terms:**\n\n\\[\n4 - 2\n\\]\n\n\\[\n2\n\\]\n\n### Step 3: Write the Simplified Expression\n\nNow, I'll combine these results:\n\n\\[\n2k + 2\n\\]\n\nSo, the simplified form of the expression \\((-k + 4) + (-2 + 3k)\\) is:\n\n\\[\n2k + 2\n\\]\n\n### Step 4: Identify the Coefficient of \\(k \\)\n\nNow The coefficient of \\(k \\) in the expression \\(2k + 2 \\) is the number multiplied by \\(k \\), which is **2**.\n\n### Final Answer\n\nTherefore, the coefficient of \\(k \\) is:\n\n\\[\n\\boxed{2}\n\\]",

Agent5: " What is the coefficient of k?\nAlright, I have this math problem to solve:\n\n**Simplify \\((-k + 4) + (-2 + 3k)\\). What is the coefficient of \\(k\\)?**\n\nFirst, I need to simplify the expression \\((-k + 4) + (-2 + 3k)\\). After simplifying, I should find the coefficient of \\(k\\), which is the number that \\(k\\) is multiplied by.\n\n### Step 1: Understand the Expression\n\nThe expression is \\((-k + 4) + (-2 + 3k)\\). I need to simplify this by combining like terms. \n\n**What are like terms?**\n\nLike terms are terms that have the same variable raised to the same power. In this case, \\(-k\\) and \\(3k\\) are like terms because they both have the variable \\(k\\) raised to the first power. Similarly, \\(4\\) and \\(-2\\) are like terms because they are both constants (they don't have any \\(k\\) in them).\n\n### Step 2: Remove the Parentheses\n\nFirst, I'll remove the parentheses to make it easier to combine the terms.\n\n\\[\n(-k + 4) + (-2 + 3k)\n\\]\n\nRemoving the parentheses (and noting that adding \\(-2\\) is the same as subtracting \\(2\\)):\n\n\\[\n-k + 4 - 2 + 3k\n\\]\n\nWait, is that correct? Let's double-check.\n\nActually, the original expression is:\n\n\\[\n(-k + 4) + (-2 + 3k)\n\\]\n\nSo, I'm adding two groups:\n\n1. \\(-k + 4\\)\n2. \\(-2 + 3k\\)\n\nWhen I add them together, it's:\n\n\\[\n(-k + 4) + (-2 + 3k) = -k + 4 - 2 + 3k\n\\]\n\nYes, that makes sense. I'm just adding the two parts together.\n\n### Step 3: Combine Like Terms\n\nNow, I'll combine the like terms. Like terms are those that contain the same variable or are constants.\n\n**Combining the \\(k\\)-terms:**\n\n\\[\n-k + 3k\n\\]\n\n\\[\n(-1 + 3)k = 2k\n\\]\n\n**Combining the constant terms:**\n\n\\[\n4 - 2\n\\]\n\n\\[\n2\n\\]\n\nSo, putting it all together:\n\n\\[\n2k + 2\n\\]\n\n### Step 4: Identify the Coefficient of \\(k\\)\n\nThe simplified expression is \\(2k + 2\\). The coefficient of \\(k\\) is the number that \\(k\\) is multiplied by. In this case, that's \\(2\\).\n\n### Verification\n\nLet's verify each step to make sure I didn't make any mistakes.\n\n1. **Original Expression:**\n\n \\[\n (-k + 4) + (-2 + 3k)\n \\]\n\n2. **Removing Parentheses:**\n\n \\[\n -k + 4 - 2 + 3k\n \\]\n\n - \\(-k\\) from the first parentheses.\n - \\(+4\\) from the first parentheses.\n - \\(+(-2)\\) from the second parentheses (which is \\(-2\\)).\n - \\(+3k\\) from the second parentheses.\n\n So, combining them:\n\n \\[\n -k + 4 - 2 + 3k\n \\]\n\n3. **Combining Like Terms:**\n\n - \\(-k + 3k = 2k \\)\n - \\(4 - 2 = 2 \\)\n\n Resulting in:\n\n \\[\n 2k + 2\n \\]\n\nEverything checks out. The coefficient of \\(k\\) is indeed \\(2\\).\n\n### Final Answer\n\nThe coefficient of \\(k\\) in the simplified expression is:\n\n\\[\n\\boxed{2}\n\\]"

Figure 7. Responses after two round debate on MATH500 with 5 agents.

Critique advantage. Recall Definition 4.3:

$$\delta_{i,t} := \mathbb{E}\Big[\beta_{i,t}^{(1)} \mid \mathcal{F}_t\Big] - m_\beta \, p_{i,t}. \tag{18}$$

E.2. one-step drift decomposition

Lemma E.1. *Under* (8)–(9) *and* $\|\beta_{i,t-1}\|_1 = m_\beta$,

$$\begin{aligned}\mathbb{E}[p_{i,t} \mid \mathcal{F}_{t-1}] = p_{i,t-1} &+ \frac{\delta_{i,t-1}}{Z_{i,t-1}} \\ &+ \frac{w_i|N(i)|}{Z_{i,t-1}} \left(\bar{p}_{N(i),t-1} - p_{i,t-1}\right),\end{aligned} \tag{19}$$

where $C_i := m_\beta + w_i|N(i)|$, $Z_{i,t-1} := \|\alpha_{i,t-1}\|_1 + C_i$, *and* $\bar{p}_{N(i),t-1} := \frac{1}{|N(i)|}\sum_{j\in N(i)} p_{j,t-1}$.

Proof. We prove (19) by explicitly expanding $p_{i,t}$ and taking conditional expectation.

Step 1: expand the numerator and denominator. From (16), for each coordinate k,

$$\alpha_{i,t}^{(k)} = \alpha_{i,t-1}^{(k)} + \beta_{i,t-1}^{(k)} + w_i c_{i,t}^{(k)}.$$

Therefore,

$$p_{i,t} = \frac{\alpha_{i,t}^{(1)}}{\|\alpha_{i,t}\|_1} = \frac{\alpha_{i,t-1}^{(1)} + \beta_{i,t-1}^{(1)} + w_i c_{i,t}^{(1)}}{\|\alpha_{i,t}\|_1}. \tag{20}$$

Step 2: simplify the ℓ_1 norm using nonnegativity. All terms in (16) are componentwise nonnegative, hence $\|\cdot\|_1$ is additive:

$$\|\alpha_{i,t}\|_1 = \|\alpha_{i,t-1}\|_1 + \|\beta_{i,t-1}\|_1 + w_i\|c_{i,t}\|_1.$$

Using (17) and (15),

$$\|\alpha_{i,t}\|_1 = \|\alpha_{i,t-1}\|_1 + m_\beta + w_i|N(i)|. \tag{21}$$

Plugging (21) into (20) yields

$$p_{i,t} = \frac{\alpha_{i,t-1}^{(1)} + \beta_{i,t-1}^{(1)} + w_i c_{i,t}^{(1)}}{\|\alpha_{i,t-1}\|_1 + m_\beta + w_i|N(i)|}. \tag{22}$$

Step 3: take conditional expectation given $\mathcal{F}_{t-1}$. Since $\|\alpha_{i,t-1}\|_1 + m_\beta + w_i|N(i)|$ is $\mathcal{F}_{t-1}$-measurable, we have

$$\mathbb{E}[p_{i,t} \mid \mathcal{F}_{t-1}] = \frac{\alpha_{i,t-1}^{(1)} + \mathbb{E}[\beta_{i,t-1}^{(1)} \mid \mathcal{F}_{t-1}] + w_i\,\mathbb{E}[c_{i,t}^{(1)} \mid \mathcal{F}_{t-1}]}{\|\alpha_{i,t-1}\|_1 + m_\beta + w_i|N(i)|}. \tag{23}$$

Step 4: compute $\mathbb{E}[c_{i,t}^{(1)} \mid \mathcal{F}_{t-1}]$. By definition (14),

$$c_{i,t}^{(1)} = \sum_{j\in N(i)} \mathbb{1}\{y_{j,t-1} = 1\}.$$

By linearity of expectation,

$$\mathbb{E}[c_{i,t}^{(1)} \mid \mathcal{F}_{t-1}] = \sum_{j\in N(i)} \mathbb{E}[\mathbb{1}\{y_{j,t-1} = 1\} \mid \mathcal{F}_{t-1}] = \sum_{j\in N(i)} \Pr(y_{j,t-1} = 1 \mid \mathcal{F}_{t-1}). \tag{24}$$

Under the DCM marginal (13), conditioned on $\alpha_{j,t-1}$ we have

$$\Pr(y_{j,t-1} = 1 \mid \alpha_{j,t-1}) = \frac{\alpha_{j,t-1}^{(1)}}{\|\alpha_{j,t-1}\|_1} = p_{j,t-1},$$

and since $\alpha_{j,t-1}$ is $\mathcal{F}_{t-1}$-measurable, $\Pr(y_{j,t-1} = 1 \mid \mathcal{F}_{t-1}) = p_{j,t-1}$. Thus (24) becomes

$$\mathbb{E}[c_{i,t}^{(1)} \mid \mathcal{F}_{t-1}] = \sum_{j\in N(i)} p_{j,t-1} = |N(i)|\,\bar{p}_{N(i),t-1}, \qquad \bar{p}_{N(i),t-1} := \frac{1}{|N(i)|}\sum_{j\in N(i)} p_{j,t-1}. \tag{25}$$

Step 5: substitute and express in terms of $\delta_{i,t-1}$. Also note that

$$\alpha_{i,t-1}^{(1)} = p_{i,t-1}\|\alpha_{i,t-1}\|_1. \tag{26}$$

Plugging (25) and (26) into (23) gives

$$\mathbb{E}[p_{i,t} \mid \mathcal{F}_{t-1}] = \frac{p_{i,t-1}\|\alpha_{i,t-1}\|_1 + \mathbb{E}[\beta_{i,t-1}^{(1)} \mid \mathcal{F}_{t-1}] + w_i|N(i)|\,\bar{p}_{N(i),t-1}}{\|\alpha_{i,t-1}\|_1 + m_\beta + w_i|N(i)|}. \tag{27}$$

Define $C_i := m_\beta + w_i|N(i)|$ and $Z_{i,t-1} := \|\alpha_{i,t-1}\|_1 + C_i$. Subtract $p_{i,t-1}$ from (27):

$$\mathbb{E}[p_{i,t} \mid \mathcal{F}_{t-1}] - p_{i,t-1} = \frac{\mathbb{E}[\beta_{i,t-1}^{(1)} \mid \mathcal{F}_{t-1}] - m_\beta p_{i,t-1}}{Z_{i,t-1}} + \frac{w_i|N(i)|(\bar{p}_{N(i),t-1} - p_{i,t-1})}{Z_{i,t-1}}. \tag{28}$$

By Definition 4.3, the first numerator is exactly $\delta_{i,t-1}$. Rearranging (28) yields (19). □